SIMATS ENGINEERING

ASSESSMENT TOOL-2

CO-1

Assignment Title: Agile Sprint Planning & User Story Workshops

AI-Powered Smart Court Case Management and Judicial Analytics Platform

COURSE: SOFTWARE ENGINEERING

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1. Introduction

The judicial system plays a vital role in maintaining justice and the rule of law in society. Every day, courts handle thousands of legal cases involving civil disputes, criminal offenses, family matters, property issues, and many other legal proceedings. As the number of cases continues to increase, traditional court management systems face significant challenges such as delays in processing, manual handling of legal documents, inefficient hearing schedules, and limited access to analytical information.

Most existing court systems rely heavily on manual documentation and administrative procedures, making it difficult to manage large volumes of case records efficiently. This often leads to increased case backlogs, scheduling conflicts, and delays in delivering justice. Additionally, judges and court administrators have limited tools to analyze judicial workloads, predict case completion timelines, or prioritize urgent cases effectively.

The rapid advancement of technologies such as Artificial Intelligence (AI), Natural Language Processing (NLP), Cloud Computing, Data Analytics, and Digital Document Management provides an opportunity to modernize judicial systems. By integrating these technologies into court operations, routine administrative tasks can be automated, legal documents can be classified intelligently, hearing schedules can be optimized, and judicial performance can be monitored using real-time analytics.

The proposed **AI-Powered Smart Court Case Management and Judicial Analytics Platform** aims to create a centralized digital platform that simplifies court operations while improving transparency, efficiency, and accessibility. The system will assist judges, lawyers, court staff, and citizens by providing secure online access to case information, intelligent document processing, automated scheduling, predictive analytics, and comprehensive dashboards for decision-making.

Ultimately, this platform contributes to faster case resolution, reduced judicial backlog, improved resource utilization, enhanced transparency, and better public trust in the judicial system.

2. Industry Context

Judicial institutions worldwide are undergoing digital transformation to improve the efficiency and transparency of legal services. Many countries are adopting e-Court initiatives that replace paper-based procedures with digital case management systems. These systems enable courts to manage legal records electronically, provide online services to citizens, and improve communication among judges, lawyers, and court staff.

Despite these advancements, many courts still experience challenges such as delayed hearings, excessive paperwork, inefficient case allocation, and limited analytical capabilities. Court administrators often spend considerable time organizing schedules and maintaining records manually, while judges have limited support for prioritizing cases based on urgency or workload.

Artificial Intelligence and Natural Language Processing have the potential to revolutionize judicial administration by automating repetitive tasks, analyzing legal documents, identifying patterns in historical case data, and predicting expected case durations. Cloud-based platforms further enhance accessibility by allowing authorized users to securely access court information from multiple locations while ensuring data integrity and security.

An AI-powered judicial management system not only improves operational efficiency but also supports data-driven decision-making, enabling governments and judicial authorities to allocate resources effectively and reduce case backlogs.

3. Problem Statement

Courts often experience delays in case processing due to manual documentation, inefficient scheduling of hearings, and limited use of analytical tools. Existing court management systems provide minimal support for intelligent case prioritization, judicial workload analysis, and predictive decision-making. As a result, legal proceedings become time-consuming, administrative workloads increase, and citizens experience delays in receiving justice.

The proposed AI-Powered Smart Court Case Management and Judicial Analytics Platform addresses these challenges by digitizing court records, automating legal document classification, optimizing hearing schedules, predicting case completion timelines, and providing real-time judicial analytics. The platform aims to improve court efficiency, enhance transparency, reduce case backlog, and ensure better access to justice for all stakeholders.

4. Project Vision:

The vision of the AI-Powered Smart Court Case Management and Judicial Analytics Platform is to transform the traditional judicial system into an intelligent, digital, and data-driven platform that improves the efficiency, transparency, and accessibility of court operations. By integrating Artificial Intelligence (AI), Natural Language Processing (NLP), Cloud Computing, and Data Analytics, the platform aims to automate routine judicial processes, reduce manual effort, and support informed decision-making.

The system is designed to assist judges, lawyers, court staff, and citizens by providing secure access to case information, automated legal document management, intelligent hearing scheduling, predictive case duration analysis, and comprehensive judicial dashboards. Ultimately, the platform seeks to reduce case backlogs, enhance the quality of judicial services, and improve public trust in the legal system.

5. Project Objectives

The primary objectives of the proposed system are:

1. To digitize court case records and judicial documents.
2. To automate legal document classification using Artificial Intelligence.
3. To predict case completion timelines based on historical case data.
4. To optimize hearing schedules and reduce scheduling conflicts.
5. To monitor and analyze judicial workload using real-time analytics.
6. To generate comprehensive legal reports and performance dashboards.
7. To improve transparency and accountability in judicial administration.
8. To reduce case backlog and improve court efficiency.
9. To provide secure online access for judges, lawyers, court staff, and citizens.
10. To support data-driven decision-making within the judicial system.

6. Stakeholder Identification

A stakeholder is any individual, group, or organization that is directly or indirectly affected by the development or use of the software system. Identifying stakeholders helps ensure that the system meets the needs of all users involved in the judicial process.

The following stakeholders are involved in the proposed system:

Stakeholder	Role in the System
Judge	Reviews case details, schedules hearings, delivers judgments, and monitors pending cases.
Court Administrator	Manages court operations, assigns cases, schedules hearings, and oversees overall court administration.
Court Clerk	Registers new cases, uploads legal documents, maintains digital records, and updates case information.
Lawyer	Files new cases, uploads supporting documents, checks hearing schedules, and tracks case progress.
Citizen/Litigant	Views case status, hearing dates, and judgment information through the online portal.
Judicial Administration Department	Monitors court performance, workload distribution, and case statistics across different courts.
System Administrator	Manages user accounts, system security, database maintenance, backups, and access permissions.
AI Analytics Engine	Automatically classifies legal documents, predicts case durations, and generates analytical insights.

7. Stakeholder Responsibilities

- **Judge**

- Review assigned cases.
- Conduct hearings.
- Deliver judgments.
- Monitor pending and completed cases.
- View AI-generated case analytics.

- **Court Administrator**

- Manage court schedules.
- Allocate cases to judges.
- Monitor hearing calendars.
- Generate court performance reports.
- Supervise court operations.

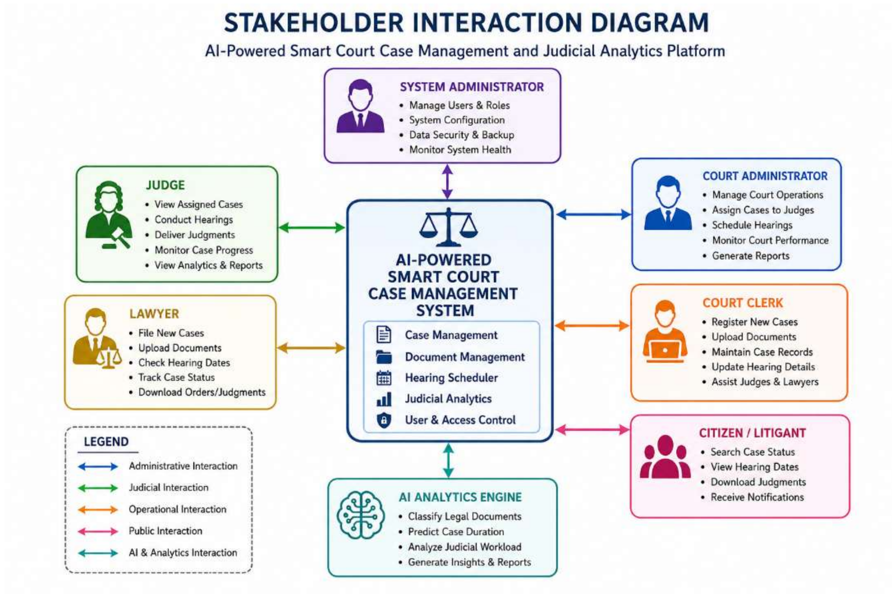
- **Court Clerk**

- Register new legal cases.
- Upload and maintain digital documents.
- Verify case information.
- Update hearing details.
- Assist judges and lawyers with documentation.

- **Lawyer**
- File new legal cases.
- Upload supporting evidence.
- Monitor hearing schedules.
- Download court orders.

- **Citizen/Litigant**
- Search for case status.
- View hearing dates.
- Download publicly available judgments.
- Receive case notifications.
- **System Administrator**
- Create user accounts.
- Manage user permissions.
- Ensure cybersecurity.
- Perform database backups.
- Maintain system availability.

8.Stakeholder Interaction Diagram:



9. Epics:

1. Epic 1: User Authentication and Access Management

This epic focuses on providing secure access to the system. Different stakeholders such as judges, lawyers, court clerks, administrators, and citizens require different levels of access based on their roles. The system should authenticate users securely and enforce role-based permissions to protect sensitive judicial information.

Main Features:

- User registration
- Secure login
- Role-based access control
- Password reset
- User profile management

2. Epic 2: Court Case Management

This epic enables efficient management of court cases by digitizing the entire case lifecycle. Lawyers can register cases, clerks can update records, and judges can review case details through a centralized digital platform.

Main Features:

- New case registration
- Case assignment
- Case updates
- Case search
- Digital case records

3. Epic 3: Legal Document Management

This epic provides facilities for securely storing and managing legal documents in digital format. It eliminates manual paperwork while improving document accessibility and reducing administrative workload.

Main Features:

- Document upload
- Document download
- Secure document storage
- Version management
- Digital document retrieval

4. Epic 4: AI-Based Legal Document Classification

This epic uses Artificial Intelligence and Natural Language Processing (NLP) to automatically classify legal documents into appropriate categories. Automation reduces manual effort, improves accuracy, and speeds up document processing.

Main Features:

- Automatic document classification
- Document tagging

- AI-powered text analysis
- Intelligent document search
- Category recommendations

5. **Epic 5: Intelligent Hearing Scheduling**

This epic automates hearing scheduling by considering judge availability, courtroom availability, and case priority. Intelligent scheduling minimizes conflicts and ensures efficient use of judicial resources.

Main Features:

- Hearing calendar
- Automatic schedule generation
- Conflict detection
- Judge availability tracking

6. **Epic 6: Public Case Tracking and Notification System**

This epic improves transparency by allowing citizens and lawyers to check case status online. The notification system keeps users informed about hearing dates, case updates, and judgments.

Main Features:

- Online case status tracking
- Hearing notifications
- Judgment notifications
- SMS/Email alerts
- Public case search

User Stories:

A User Story is a short and simple description of a software feature from the perspective of the end user. It captures what the user wants to achieve and the value it provides. In Agile development, user stories help developers understand user requirements and prioritize features during sprint planning.

The standard format of a user story is:

As a <User>, I want <Feature>, so that <Benefit>.

ID	**User Story**
US-01	As a **System Administrator**, I want to create user accounts, so that authorized users can access the system.
US-02	As a **User**, I want to log in securely, so that my information is protected.
US-03	As a **Lawyer**, I want to register a new case, so that it is recorded in the system.
US-04	As a **Court Clerk**, I want to upload legal documents, so that case records are stored digitally.
US-05	As a **Judge**, I want to view assigned cases, so that I can prepare for hearings.
US-06	As a **Court Administrator**, I want to schedule hearings automatically, so that scheduling conflicts are reduced.
US-07	As a **Court Clerk**, I want AI to classify legal documents, so that manual work is minimized.
US-08	As a **Judge**, I want to view judicial analytics, so that I can prioritize important cases.
US-09	As a **Judge**, I want AI to predict case completion time, so that hearings can be planned effectively.

Sprint Backlog:

Task	**Assigned To**
Develop Login Module	Developer

Task	Assigned To
Create Case Registration	Developer
Build Document Upload	Developer
Integrate AI Classification	AI Engineer
Test Features	QA Tester
Fix Bugs	Development Team

Sprint Goal

Develop the core modules of the Smart Court Case Management System, including user authentication, case registration, document upload, and AI-based document classification.

Expected Deliverables

- User Login Module
- Case Registration Module
- Document Upload Feature
- AI Document Classification
- Basic Database

Conclusion:

The AI-Powered Smart Court Case Management and Judicial Analytics Platform will improve the efficiency of judicial processes by digitizing case management, automating document classification, optimizing hearing schedules, and providing AI-based analytics. The Agile sprint planning approach enables the project to be developed incrementally while ensuring continuous improvement and timely delivery of features.